BENJAMIN BARLOG

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EDUCATION

University of Montana, Missoula, MT — Expected May 2027

B.S. Computer Science, Data Science concentration; Minor in Mathematics · GPA: 3.61 (institution)

Relevant coursework: Data Structures & Algorithms, Operating Systems, Computer Architecture, Probability & Simulation, Linear Algebra, Discrete Mathematics, Data Visualization, Geospatial Data Science, Machine Learning

RESEARCH EXPERIENCE

Undergraduate Research Assistant, AI in Education — University of Montana — Jan 2026 – Present

Mentor: Dr. Zedong Peng, Department of Computer Science

Studied whether LLM tutors give responses that suit a learner's knowledge and place in a course, not just correct ones. Developed the Pedagogical Suitability Index (PSI), a composite metric for scoring the pedagogical fit of tutoring responses, and used it to evaluate and improve the output of four LLMs.

- Designed the PSI evaluation metrics: temporal violation, knowledge distance, cognitive load, and Bloom's taxonomy alignment
- Applied PSI across 240 evaluations of four LLMs; PSI-guided regeneration, built by collaborators, improved 51 of 62 low-scoring responses (82.3%)
- First author; presented the paper at IEEE IRI 2026 in Seattle, WA

PUBLICATIONS & PRESENTATIONS

B. Barlog, H. Craig, and Z. Peng, "Evaluating and Improving Pedagogical Fit in LLM-Based AI Tutors with the Pedagogical Suitability Index," in *Proc. IEEE 27th Int. Conf. on Information Reuse and Integration for Data Science (IRI)*, Seattle, WA, USA, 2026. arXiv:2608.05411.

- Paper presentation, IEEE IRI 2026, Seattle, WA, Jul. 31 – Aug. 2, 2026

TECHNICAL PROJECTS

GNSS Interference Hotspots — Python — Oct 2026 – Present

Geospatial Data Science course project (in progress) · Instructor: Dr. Jesse Johnson

Testing where GPS interference is spatially clustered, how persistent those clusters are, and whether they hold up after accounting for aircraft traffic density and receiver coverage, using daily ADS-B-derived H3 hexagon data (GPSJAM) for the Baltic region.

- Building a spatial-autocorrelation pipeline: contiguity weights from H3 hexagon topology, global Moran's I time series, and LISA hotspot maps with false discovery rate correction
- Using Empirical Bayes rate smoothing and minimum-sample thresholds so low-traffic cells do not produce false hotspots

Terrain Masking and Least-Exposure Routing — Python, Observable — Oct 2026 – Present

Data Visualization course project (in progress) · Instructor: Dr. Jesse Johnson

Testing whether least-exposure routes for low-altitude drones under notional line-of-sight sensors favor lower, more rugged terrain than shortest routes, and how that changes with altitude. Sole author of the hypothesis, method, and implementation.

- Building a pipeline on USGS 3DEP elevation data: per-sensor minimum-visible-altitude rasters, probabilistic detection, and a survival-based path cost ($-\ln(1-P)$) solved exactly with Dijkstra's algorithm
- Designed a paired statistical test (Wilcoxon signed-rank) across randomized sensor layouts, start–goal pairs, and three flight altitudes

Ultrasonic Scanning Sonar — Arduino, C++, Python — Sep 2026

- Built a servo-swept ultrasonic scanner (Arduino Uno, HC-SR04) that streams angle–distance readings over serial to a live polar plot, mapping a 180° slice of a room
- Wrote non-blocking servo timing and 6-sample per-angle averaging to reduce noise; detected a test object at 25 cm

Housing Affordability in Recreation-Amenity Counties — Python, QGIS Aug – Sep 2026

Geospatial Data Science course project · Instructor: Dr. Jesse Johnson

Tested whether counties with strong outdoor-recreation access and little high-wage industry are unaffordable for local workers, across 2,753 U.S. counties.

- Built a home-price-to-local-wage metric (Zillow, BLS QCEW) and a recreation amenity index from federal-land polygon overlays and recreation employment (GeoPandas, statsmodels, QGIS)
- Found a significant amenity × industry interaction ($p < 10^{-13}$): amenity's effect on unaffordability was ~3× larger where high-wage industry was scarce, acting through prices rather than wages
- Caught and corrected a boundary-data error that overstated eastern national-forest land by up to 55 percentage points

Swarm Intelligence Self-Study — Python Aug 2026

- Directed an AI-assisted build of an interactive 2D drone-swarm simulator (boids flocking, noisy range-limited sensing, Kalman filtering) and used it to study how individual parameters and sensor noise affect swarm cohesion

Data Visualization Studies — Observable 2026

- Coastal proximity and fitness: coastal counties show more fitness-related activity, likely reflecting affluence rather than the coastline itself
- Campaign finance and wealth inequality: sparse donation data supported at most a weak positive correlation

PROFESSIONAL EXPERIENCE

Personacue.ai — Co-founder & Lead Backend Developer Mar 2026 – Present

Mentor: Dr. Zedong Peng, University of Montana; with Hudson Craig (lead frontend)

LLM-based platform that replicates the personalities of historical and fictional figures on locally hosted models. Details are limited by a non-disclosure agreement.

- Lead backend development; co-developed the original retrieval-augmented generation (RAG) pipeline for aligning local models with personas, roles, and knowledge bases

First Interstate Bank — Database Administrator Intern, Billings, MT (remote) Jun 2025 – Sep 2025

- Developed an in-house server-monitoring solution to replace a third-party system, saving roughly $70,000 per year
- Supported DBAs with server health monitoring, incident reporting, deadlock troubleshooting, and query optimization (SQL Server, SSMS, SolarWinds DPA, Power BI)

First Interstate Bank — Wealth Management Admin. Assistant, Billings, MT Jul 2022 – Dec 2022

- Scheduled appointments, initiated transactions, and distributed RMDs
- Prepared and delivered presentations on current economic conditions to colleagues
- Served as a liaison between clients and advisors

SKILLS

Languages & Frameworks: Python, C/C++, Java, SQL, JavaScript, Django
Data & ML: pandas, NumPy, GeoPandas, statsmodels, matplotlib, QGIS, Observable
Tools: Windows, Linux, Git, Visual Studio, Jupyter, PlatformIO/Arduino, SQL Server (SSMS)
Interests: economics, autonomy, fitness, investing, poker, cooking
Citizenship: U.S. Citizen; eligible for security clearance